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Long-Term Effects of Strategic Reading Instruction in the Intermediate Elementary Grades

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ABSTRACT

The purpose of the present study was to examine the effects of a program that offered sustained strategic reading instruction on reading abilities of third and fourth graders. The study was conducted among 1,469 children from 40 schools in the Netherlands. Schools were randomly assigned to either the experimental or control group. Multilevel analyses showed positive effects on knowledge of reading strategies after one year of intervention at the end of Grade 3. At the end of Grade 4, there were also positive intervention effects on reading comprehension. No significant interaction effects were determined for age, gender, SES, or ethnic/linguistic background. The intervention effects were also not dependent on the level of the student's nonverbal intelligence, vocabulary, or decoding skills.

KEYWORDS

reading comprehension
reading strategies
intervention study

Introduction

The importance of the acquisition of good reading comprehension is widely acknowledged. Reading comprehension is essential to student success in school and full participation in society. There is much concern about how to prepare students to independently read and comprehend increasingly complex texts in Grade 3 and beyond. Since the 1980s, the instruction of reading strategies has received great attention. Although the debate on whether strategy instruction is the best way to teach comprehension skills is ongoing (e.g., McKeown, Beck, & Blake, 2009), many well-controlled experimental studies have shown that children's reading comprehension can be fostered through strategy instruction (e.g., Palincsar & Brown, 1984; Rosenshine & Meister, 1994; Spörer, Brunstein, & Kieschke, 2009; van Keer, 2005). In most of these studies, the intervention period was limited and in only a few studies were the effects of multiple strategies examined (e.g., Reutzel, Smith, & Fawson, 2005; Spörer et al., 2009). Thus, because a more sustained instruction perspective in natural classrooms is lacking (Andreassen & Bråten, 2011), the aim of the present study was to examine the sustained effects of a program for strategic reading instruction (SRI) over a period of two years in a natural classroom setting.

Research among proficient readers shows that good readers use effective strategies to better understand text (Pressley & Afflerbach, 1995). Thus, the teaching of effective reading

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strategies is important for helping children to improve comprehension and learning from text (McNamara, 2007). Since the 1980s, many studies have examined the effects of systematic instruction in reading strategies on reading comprehension. Palincsar and Brown (1984) were the first to show that the reciprocal teaching of four strategies—predicting, clarifying, questioning, and summarizing—improved reading comprehension results of struggling readers. Many researchers confirmed this finding. In two meta-analyses, consistent mean effect sizes of 0.32 (Rosenshine & Meister, 1994) and 0.31 (Davis, 2010) on standardized reading comprehension measures were found in experimental studies using reciprocal teaching in different contexts and with different age groups. Other well-examined approaches that proved to have significant effects on students' reading comprehension are, among others, transactional strategy instruction (Brown, Pressley, van Meter, & Schuder, 1996; Pressley, El-Dinary, Wharton-McDonald, & Brown, 1998) and concept-oriented reading instruction (Guthrie, 2003).

Although it is not clear exactly which strategies must be taught, there is consensus that a limited set of multiple strategies is better than an exhaustive amount of strategic activities (Davis, 2010; National Reading Panel, 2000; RAND Reading Study Group, 2002). Based on an extensive literature review, Block and Duffy (2008) identified nine strategies that are most effective: predicting, monitoring, questioning, creating mental images, using repair strategies such as rereading, inferring, summarizing, evaluating, and synthesizing. These strategies are consistent with the report of the National Reading Panel (NRP, 2000) that identified seven strategies as having a solid evidence base: comprehension monitoring, graphic and semantic organizers, questioning, text structuring, summarizing, cooperative learning, and multiple-strategy instruction. The meta-analysis of Davis (2010) found that interventions that included previewing the text before reading, creating graphic organizers, and providing time for students to analyze and reflect on their reading produced larger effect sizes on measures for strategy knowledge and strategy use (as measured, for instance, in thinking aloud) and on measures for reading comprehension compared to studies that did not include these strategies.

Strategy instruction is theoretically grounded in theories of self-regulated learning (Butler & Winne, 1995; Zimmerman, 2002). Self-regulated learners have been found to successfully use cognitive activities, to control and monitor their learning with meta-cognitive activities (such as orientation, planning, monitoring, and evaluating) and to motivate themselves (Zimmerman, 2002). It can be assumed that to effectively use reading strategies, students not only need to build explicit, declarative knowledge about strategies but must also acquire the appropriate self-regulation skills. It is not enough for students to know about strategies and be familiar with their functions. They must learn to adapt and accommodate the strategies to a specific reading situation. This implies that students should become aware of their own repertoire of reading strategies to choose the appropriate one and then evaluate whether the strategy led to the expected goal (De Corte, Verschaffel, & Van de Ven, 2001). Consistent with the theory of cognitive apprenticeship, the teacher may serve as a model in demonstrating when and how to apply strategies while reading text (Collins, Brown, & Newman, 1989).

However, teachers often find it challenging to effectively teach and model for students how to use comprehension strategies, and many researchers have noticed that the implementation of strategy instruction could be improved. Although teachers and students often use the terminology of strategy instruction (i.e., they talk about “making connections” and “activating background knowledge”), they find it difficult to apply strategy instruction

regularly in their classrooms (Anmarkrud & Bråten, 2012; Davis, 2010; Pressley, 2000, Souvignier & Mokhlesgerami, 2006). It has also been noticed that teachers using existing programs often change or leave out essential elements of strategy instruction (Aarnoutse & Schellings, 2003; Souvignier & Mokhlesgerami, 2006). Manuals developed by educational publishers often describe activities, but they do not provide enough explicit clues for modeling instruction or for guided and independent practice by the students (Block & Duffy, 2008). Moreover, many materials offer a vast amount of strategic activities that can only be exercised once or twice during a school year (Block & Duffy, 2008; Stoeldraijer & Verhooy, 2007).

To conclude, in the research thus far, there is evidence of immediate effects of strategy instruction in controlled settings. However, it is by no means clear to what extent strategy instruction can be implemented on a wider scale or to what extent this may lead to sustained effects in the explicit knowledge of reading strategies, on the one hand, and reading comprehension skills, on the other hand. Nor is it clear which student and background variables account for such effects. To address these topics, the present study examines the long-term effects of strategy instruction using a large sample of intermediate elementary classrooms in the Netherlands.

The Present Study

In the present study, we evaluated the effects of an SRI program that aims to help teachers conduct instruction on a limited set of strategies in a meaningful context. The core of the SRI program consists of teacher instruction that involves demonstrating and modeling strategies in the meaningful context of reading aloud interesting text. This is then followed by the students applying the modeled strategies while they read books of their own choice during independent reading. As a scaffolding and reflection aid, the students used a strategy checklist on which they noted their problems and solutions. The same strategies were taught and deepened over a two-year period while emphasizing student self-regulation. Scharlach (2008) showed that reading aloud children's books may offer a rich and meaningful context to model the strategies. Strategies modeled during read-aloud sessions may be applied during independent reading and transferred to content area reading. Furthermore, it takes time to become a strategic self-regulated reader because children may acquire the declarative knowledge about strategies in a limited period, but to learn to apply the strategies effectively when appropriate and necessary requires time and effort. Therefore, strategy instruction should be sustained and should gradually shift from external regulation by the teacher to internal regulation by the students. Accordingly, our program was implemented over a two-year period in Grades 3 and 4.

To examine the effects of the SRI program, which is described in detail in the method section, a study was set up using a pretest/posttest control group design. A group of 40 schools participated in the study for two years. The schools were randomly assigned to either the experimental group or the control group. At the beginning of the first year, the third-grade teachers of the experimental group received a short training on how to implement the program. This training was repeated for the fourth-grade teachers at the beginning of the second year. The implementation of the program in classrooms was monitored through observations. The large sample made it possible to take into account the hierarchically nested design of students in classes in conducting multilevel modeling. In addition, as we were able to

follow the same group of students for two years, thereby reducing novelty effects, it was possible to examine the sustained effects of the program on both strategy knowledge and reading comprehension. The following research questions were addressed:

1. What is the effect of the SRI program on the knowledge of reading strategies and on reading comprehension after one year and after two years of intervention?
2. Are there differential effects for children due to age, SES, ethnic/home language background, decoding skills, vocabulary, or nonverbal IQ?

Method

Design

To examine the effects of the SRI program, a pretest/posttest control group design was used. Schools were randomly assigned to either the experimental or the control condition. Students in the experimental group received instruction based on the SRI program for two school years in Grades 3 and 4, while the control group received instruction using the regular school reading curriculum. Measurements occurred at three moments in time: the pretest at the beginning of third grade (September 2009), the intermediate posttest at the end of third grade (June 2010) and the final posttest at the end of fourth grade (June 2011).

Recruitment and Selection of Schools

The study was conducted in cooperation with a Dutch center for school counseling, HCO, located in The Hague. All 222 schools in the region of this center were invited, through a written letter, to participate in the study. Within three weeks, 45 schools had agreed to participate. These schools were located in The Hague and surrounding cities. At an informational meeting for school principals, the research procedures and the expectations of the school were explained. The randomization procedure, the testing load, and the types of tests to be administered were discussed. In addition, it was made clear that the experimental schools were not allowed to use their own reading comprehension curriculum for the duration of the study and that the regular program was to be completely replaced by the experimental program. None of the schools withdrew after this meeting.

We randomly assigned schools rather than classrooms/teachers to the two treatments to prevent treatment diffusion within schools. Because of the length of the intervention, cooperation between teachers was likely. An independent researcher from Radboud University performed the randomization.

All schools were informed about their group assignment during a school visit just prior to the summer holidays. Despite our efforts to keep all schools on track, five schools withdrew at the beginning of the study (four control and one experimental), due primarily to internal management problems within the schools. Just prior to the final posttest one experimental school and one control school withdrew because they were unable to organize the testing due to changes in location, that is, moving to different buildings. The available data of the last two schools were used in the analyses. We had, however, no data from the five schools that dropped out already at the beginning; this is a relatively low attrition rate but differential attrition was high (18% of the

control schools and 4.3% of the intervention schools). This differential attrition may have contributed to pretest differences between the two conditions, because three of the four nonparticipating control schools were from low-SES neighborhoods and had relatively high percentages of students from ethnic-minority backgrounds. Thus, differential attrition may explain the pretest differences: the control group scored higher at pretest than the intervention group, but these pretest differences disappeared when background variables at the classroom level were controlled for (see Appendix C in the online supplemental material).

In total, 40 schools with 62 classrooms participated both in Grade 3 and Grade 4. The experimental group comprised 22 schools with 34 classrooms, while the control group included 18 schools with 28 classrooms. The teachers of Grade 3 ($n = 62$) and Grade 4 ($n = 62$) all had the legally required bachelor's degree obtained at one of the accredited teacher training institutes. All teachers were experienced teachers, with at least five years of experience.

Participating Students

Of the 1,469 students participating, 842 were enrolled in the experimental schools and 627 attended the control schools. In total, 1,383 students were administered the pretest. Because of the longitudinal design, the student body of the participating schools changed during the study. After the pretest, 86 new students entered the participating schools, and during the two school years, 164 students transferred to other schools, mainly because they moved to another neighborhood. As a result, the number of students differs for each of the three measurement points. See Appendix C in the online supplemental material for an analysis of the missing values and the pretest differences between experimental conditions. Both overall attrition and differential attrition at the student level stayed within the limits of acceptability as defined in the *Procedures and Standards Handbook Version 3.0* (What Works Clearinghouse, 2013).

Demographic information collected for each student included date of birth, gender, SES, ethnicity, and home language. Our indicator for SES, which was solely based on the level of education obtained by the parents, was a three-category variable based on the weight factors the Dutch government uses for school funding to compensate for disadvantaged students. We denoted the three categories as follows: “low” if one of the parents had only primary education and/or the lowest level of special education, “lower middle” if both parents had finished the lowest levels of secondary education (the lower streams of vocational education), and “not disadvantaged” if one of the parents finished secondary education or higher. As ethnicity of the children was defined by the birth country of the parents, a dichotomous variable was constructed: both parents born in the Netherlands versus one or both parents born in another country. In addition, we asked the students about their home language (HL). The answers of the children were categorized into three groups: mostly Dutch, two languages equally, or mostly a language other than Dutch. This information was combined with ethnicity into a variable with three categories: (a) Dutch/HL Dutch; (b) non-Dutch/HL mixed (meaning Dutch or two languages equally at home); (c) non-Dutch/HL non-Dutch. Table 1 presents an overview of the student characteristics separately for the two experimental conditions. Boys and girls were fairly equally represented in both conditions. However, there were

Table 1. Overview of student characteristics.

	Condition	
	Control group <i>N</i> = 627	Experimental group <i>N</i> = 842
Gender		
Boy	329	415
Girl	298	427
Mean age (at pretest)	8;7	8;8
SES		
Not disadvantaged	468	551
Lower middle	55	72
Low	95	201
	(<i>N</i> = 618)	(<i>N</i> = 824)
Ethnicity and Home Language		
Dutch	319	343
Mixed	117	297
Non-Dutch	91	201
	(<i>N</i> = 527)	(<i>N</i> = 841)

Note. For explanation of the variables, see text.

some missing values for SES, ethnicity, and HL (see Appendix C in the online supplemental material).

The Strategic Reading Instruction Program

The SRI program draws on the theoretical framework of self-regulated learning and cognitive apprenticeship (see the introduction to this article). The main aim of the program is to create independent self-regulated readers who are capable of applying appropriate reading strategies in various contexts. In developing our program, we used the instructional framework developed by Scharlach (2008). The intervention consisted of reading comprehension instruction integrated with reading aloud and independent reading (three times a week). It also included integrating the reading comprehension instruction once a week in other content areas such as science. The teacher offers a limited set of strategies in a meaningful context over an extended period of time. We based the choice of the particular strategies on previous research and reviews on reading comprehension instruction, in particular Block and Duffy (2008), National Reading Panel (2000), and RAND Reading Study Group (2002) (see the introduction to the present article). These reviews make it clear that multiple strategy instruction is more effective than a single strategy. There is substantial overlap in the strategies mentioned in these reviews suggesting that strategies should help orient students' attention toward the reading task by making predictions and activating prior knowledge. The reviews also highlight the importance of comprehension monitoring and active processing by means of (a) constructing mental images that represent ideas in text, (b) asking questions and seeking answers, and (c) solving comprehension problems. Finally, strategies are mentioned that help structure the information after reading a text such as constructing summaries and evaluating what has been read in order to distinguish main aspects and details and to judge the relevance of information. The selected SRI strategies were as follows—*before reading*: (a) predicting/previewing text, (b) relating text to own experience or background knowledge; *during reading*: (c) monitoring

comprehension, (d) deriving word meaning from context, (e) visualizing, (f) questioning; *after reading*: (g) generating main idea, (h) summarizing, (i) evaluating content and sharing opinions.

Structure of Read-Aloud and Independent Reading Lessons

The read-aloud was conducted three times a week. The time schedule of each lesson was as follows: (a) introduction of 5 minutes, (b) 15 minutes reading aloud including 5 minutes modeling of one or two strategies to the whole class, (c) 15 minutes individual independent reading with application of strategies and making notes, and (d) 5 minutes whole-class evaluation. The SRI lessons were given three times a week over a period of 30 weeks in each of the two school years.

Read-Aloud. Well-written, interesting children's literature by award-nominated Dutch authors was selected for the read-aloud lessons (some examples of authors and book titles: Tosca Menten, *Juffrouw Pots*; Mirjam Oldenhave, *Klem!*, and Lydia Rood, *Overleven op drakeneiland*). The books had high appeal to a wide population of students, irrespective of their reading levels or socioeconomic and cultural backgrounds. For each book, a manual was developed for the read-aloud session whereby two or three possible moments to model a strategy were highlighted. Moreover, instructions on how to model the strategy were also provided. Teachers not only modeled the implementation of the strategy, but they also explicitly explained why they used it. Teachers could use these examples literally or adapt them to their own situations and experiences. The read-aloud created a natural, meaningful, and motivating context to not only model a strategy but to also discuss solving difficulties associated with the content of the text. In Appendix A of the online supplemental material, example sentences of the modeling per strategy are given.

Independent Reading. During independent reading, the students were allowed to read books of their own choice and at their own reading level from the classroom or school library (informational books or fiction). The students used a strategy checklist designed after Scharlach (2008) to write down their problems and solutions (the result of applying the specific strategy). The students kept all checklists in a folder. During independent reading, the teacher had the opportunity to read with poorer readers and assist them as they applied the strategies. For independent reading the schools used their own collection of books. In all classrooms a collection of narrative and informative books was available. These were age-appropriate books written for children. There were also graded book series with leveled texts. In the Netherlands, one method for leveling texts (AVI-system) is mainly used by teachers. Children were free to choose from the collection and could read a book over a longer period of time.

Evaluation and Reflection. After the independent reading, the students shared their findings and reflected on the strategy they used with the teacher. The teacher discussed with the children their findings, reflected with them on their ability to apply the strategy, and discussed whether the strategy helped them to better understand the text.

Transfer to Reading Informational Texts in Content Areas

Once a week the same strategies were applied when reading text from other content areas. During these lessons, the children read a text selected by the teacher. The students individually read the text while applying strategies before and during reading. The strategy checklist functioned as a reminder of the complete strategy set. After reading, the students worked in groups of four on an assignment to create a word web of their knowledge of the topic and a summary or graphic organizer of the content of the text.

Differences Between the Programs in Grade 3 and Grade 4

In Grade 3, the strategies were introduced one by one over a nine-week period. Each week, a new strategy was introduced while the former one was repeated. After nine weeks, and after all strategies had been demonstrated and explained, the teachers modeled two strategies at a time in varying combinations. In Grade 4, the SRI was continued during read-alouds and independent readings. However, rather than focusing on only two strategies, three strategies were modeled and practiced at a time. Moreover, the strategy “visualizing” focused less on “making a picture in your head,” as it did in Grade 3, and more on creating a graphic representation of the text (i.e., displaying important information and relationships in a word web or diagram).

The ultimate goal of SRI was for the students to become independent and competent readers. Therefore, in Grade 3 the teacher modeled all strategies, while in Grade 4 the role of the teacher as a model gradually shifted to the students as they were frequently asked to assist in the modeling of the strategies. For instance, when preparing a summary, the teacher and students together determined the information that was considered more important.

The SRI strategies were aimed at developing the cognitive and metacognitive skills the students needed to become aware of and solve comprehension problems and to understand higher level text. Students of varying reading levels could practice the strategies while reading books of their own choice and at their own reading level.

Implementation of the SRI Program

The teachers of the experimental group received training to familiarize them with the intervention. This training was first given to the teachers of Grade 3 and a year later to the teachers of Grade 4. The training consisted of two meetings of three hours each in September, one meeting halfway through the school year (at the end of January) and an evaluation in June. During the meetings in September, the background and the principles of the SRI program were explained and all materials as well as the planning and organization of the lessons were discussed. There was significant focus on how teachers could model the reading strategies. We used short video clips in which one of the developers demonstrated the application of each strategy in a classroom. These clips offered a rich example and opportunity for the teachers to reflect on and to practice the modeling in small groups.

During the meeting in January, information about the SRI was refreshed, and the teachers shared experiences and problems they experienced implementing the program, most of which were related to organization and time management. The problems were discussed and solved together. They also brought folders with student material to reflect on the way students handled the strategy checklist. The final meeting was used to evaluate the intervention and to discuss the experiences of the teachers. The teachers reported that the training

and the explicit manual had been very helpful in conducting the lessons. Because the instruction was scripted they knew exactly what to do. The teachers were very enthusiastic about the books chosen to read aloud. They unanimously mentioned that the children were highly engaged during the reading aloud and that this motivated them to practice the strategies. They reported to have had more meaningful interactions about the content of books and stories. They also reported the use of the same set of strategies in other reading contexts. In the beginning of the school year teachers complained about time management and had difficulty fitting the lessons in forty minutes, but during the year they got accustomed to the new structure.

The teachers began the intervention in the beginning of October, which was after the pretest had been administered and after the second meeting on implementation procedures. In both Grade 3 and Grade 4, the teachers were observed in November to determine whether they understood the intervention and were conducting the lessons as intended. Each observation was then followed by a coaching session. We conducted this observation at an early stage to ensure that all teachers were implementing the intervention properly and to provide the teachers with feedback. The coaching was performed by three certified and experienced coaches under supervision of the second author, a certified and experienced school advisor, who wrote the manual for the teachers. The main aim of this coaching was to provide the teachers with individual feedback. The aspects that were discussed were dependent on the strength or weaknesses of the lessons. Time management and structure of the lesson, modeling of strategies, engagement of the students, and providing feedback to students were frequently discussed. The teachers also had the opportunity to ask questions and discuss problems. In most cases, we observed that the lessons were conducted well and that the teachers were implementing the lessons as described in the manual.

Control Condition

The schools in the control group used a variety of programs for reading comprehension. These programs offered two lessons per week of 45 minutes for 30 weeks and twice a week for 30 minutes for decoding skills. The programs typically involved short texts (expository and fictional), followed by a series of questions to which the students had to respond in writing. In Appendix B of the online supplement to this article we give a description of the most frequently used program.

In general, the curricula used in the control schools differed on a number of important aspects from the SRI curriculum: First, in the control group, students all read the same short texts written especially for the reading comprehension lessons. A meaningful context was lacking. After the instruction of the teacher, the students answered a number of questions in writing about the content of the text. In the SRI program, the modeling of strategies took place during reading aloud of an inspiring book by the teacher. Immediately afterward, the students practiced these strategies during independent reading with books of their own choice and on their own reading level. The children did not answer written questions, but they wrote on a student recording sheet their comments about their strategy use and their solutions. Second, in the manuals of the control group there were no clues for the teachers on how to model or explain a strategy. The SRI program provided detailed scripts for the teacher on when and how to model a strategy, what to say exactly, and which questions to ask during reading aloud.

Time Spent on Reading Comprehension Instruction

The regular time spent on reading comprehension instruction in the control groups was two lessons of 45 minutes per week over a period of 30 weeks and in addition two lessons focused on decoding skills. In the experimental group three lessons per week of 40 minutes with reading aloud and modeling of strategies were given over a period of 30 weeks. In these lessons independent reading and reading aloud were integrated. So, the amount of time spent on reading comprehension instruction in the experimental and control group was roughly the same: 90 minutes per week.

Observations of Lessons in Experimental and Control Groups

To monitor treatment implementation, all experimental and control classrooms were observed once a year. In both school years, observations took place in January/February and were conducted according to a predominant activity sampling method with a 30-second interval (Van Elsäcker, 2002). The observations were performed by the second author and three assistants. The assistants received training in which the coding scheme was explained and practiced until agreement was reached. The observers observed for 45 minutes and coded the activity of the teacher at the end of each 30-second interval. Three categories were coded: grouping, mode of activity, and content of activity. Mode of activity was coded according to whether the teacher was talking, listening, writing, or reading to the children. For content of activity, we distinguished whether the instruction was focused on content knowledge, explaining meaning of the text, instruction of strategy, modeling of strategy, discussion of assignment, evaluation of use of strategies, procedural remarks, or behavioral remarks. The observations showed some differences in instructional behavior between the experimental group and the control group. As intended, modeling of strategies occurred more frequently in the experimental than in the control group. Moreover, the teachers evaluated the use of strategies more frequently in the experimental group. The teachers in the control group, on the other hand, spent more time discussing text assignments.

In addition, at the end of a lesson, the observer rated the quality of the lesson on several aspects on a four-point scale: 1 = not present or very weak, 2 = weak, 3 = sufficient, 4 = very well. The aspects rated were: Explaining lesson goals, modeling of strategies, feedback on use of strategies, lesson evaluations, classroom management, and clarity of instruction. The data is summarized in Table 2. In Grade 3, there were statistically significant differences between the two conditions: the experimental group showed a higher level of explaining the goals of the lesson, more modeling of strategies, a higher level of feedback on the use of strategies, and more reflection and evaluation at the end of the lesson. No differences were observed with respect to classroom management or clarity of instruction.

In addition, in the SRI classrooms, specific characteristics of the program were rated on a four-point scale. It was observed that all stages of SRI were applied, i.e., reading aloud, modeling of strategies, independent reading, and reflection ($M_{\text{exp}} = 3.33$, $SD = 0.76$). Reading aloud was performed in an engaging way ($M_{\text{exp}} = 3.61$, $SD = 0.55$), classroom management during the independent reading was judged positively ($M_{\text{exp}} = 3.48$, $SD = 0.67$), and the schools had a wide variety of books of different genres available for independent reading ($M_{\text{exp}} = 3.70$, $SD = 0.58$).

Table 2. Differences in instructional behavior observed between conditions: Means, standard deviations and statistical tests.

	Control		Experimental		<i>t</i>	<i>df</i>	<i>P</i>
	<i>Mean</i>	<i>SD</i>	<i>Mean</i>	<i>SD</i>			
Explaining lesson goals							
Grade 3	1.85	1.41	2.85	1.41	−2.72	58	.009
Grade 4	2.92	1.41	2.97	1.12	−0.14	53	.89
Modeling of strategies							
Grade 3	1.30	1.56	3.03	1.01	−5.18	58	<.001
Grade 4	1.08	1.63	3.07	0.98	−5.58	53	<.001
Feedback on strategy use							
Grade 3	2.52	1.31	3.15	1.03	−2.09	58	.041
Grade 4	1.60	1.61	2.83	0.98	−3.49	53	<.001
Evaluation end of lesson							
Grade 3	1.61	1.69	2.67	1.27	−2.76	58	.008
Grade 4	1.56	1.83	3.43	1.16	−4.61	53	<.001
Classroom management							
Grade 3	3.63	.69	3.61	0.60	0.16	58	.87
Grade 4	3.13	.85	3.53	0.73	−1.75	53	.09
Clarity of instruction							
Grade 3	2.58	1.03	3.21	0.78	−1.54	58	.13
Grade 4	3.00	1.10	3.00	0.64	0.00	53	1.00

Note. Observations were rated on a scale from 1 to 4.

In Grade 4, similar patterns were found. There were clear differences in the modeling of strategies, the feedback provided regarding the use of strategies and reflection, and evaluation at the end of the lesson. No differences were observed with respect to the explanation of the goals of the lesson, the management of classroom behavior, or the clarity of instruction. Moreover, the teachers received positive scores regarding specific aspects of the SRI program. During the lessons all stages of the program were observed, i.e., reading aloud, modeling of strategies, independent reading, and reflection ($M_{\text{exp}} = 3.37$, $SD = 0.89$). Reading aloud was performed in an engaging way ($M_{\text{exp}} = 3.20$, $SD = 0.96$), and classroom management during independent reading was judged positively ($M_{\text{exp}} = 3.34$, $SD = 0.82$).

Materials

To measure the effects of the SRI program we selected general instruments that were either standardized or used in previous research.

Knowledge of Reading Strategies

Knowledge of reading strategies was measured using a questionnaire based on Brand-Gruwel (1995) with 25 multiple-choice questions. The questions were centered around solving a possible reading problem before, during, or after reading a text (e.g., you encounter a word with which you are unfamiliar). Each question had three response options: the most adequate strategy (awarded 2 points), a less adequate strategy (awarded 1 point), and an irrelevant strategy (awarded 0 points). The maximum score was 50. The reliability of the scale was .67 (Cronbach's alpha).

Reading Comprehension

Reading Comprehension Test. The Reading Comprehension Test was developed by Aarnoutse and Schellings (2003) for third and fourth graders and consisted of short narrative and expository texts followed by multiple-choice questions that drew on the respondent's ability to make inferences, identify text type and text structure, and determine main ideas. The original test consisted of 14 texts and 45 questions. We used a shortened version, however, with 11 texts and 29 questions (maximum score = 29). It took the children 45 to 60 minutes to complete the test. The reliability of the shortened test was .76 (Cronbach's alpha).

PIRLS Reading Literacy. The PIRLS test was developed for the Progress in International Reading Literacy Study (Mullis, Martin, Kennedy, & Foy, 2007), for nine- and ten-year-old students in forty countries. We used one booklet from the test battery of the PIRLS-2006. The booklet we used contained both a narrative text (*An Unbelievable Night*) and an expository text (*Searching for Food*). Half of the questions were multiple choice and the other half were open-ended questions. The open-ended questions only needed short written responses. The open-ended and multiple-choice questions both had literal and inferences questions. The questions were designed to assess the child's ability to focus on and retrieve explicitly stated information; make straightforward inferences; interpret and integrate ideas and information; and examine and evaluate content, language, and textual elements. The test was administered in two sessions on two consecutive days: first the narrative text, then the expository text. The students wrote their answers in the booklet containing the questions. The open-ended questions were scored by student assistants who were trained by the coordinator of PIRLS-Netherlands. During the training the students practiced scoring of the items by means of example answers. Spelling and grammatical errors were not taken into account. The assistants were trained not only to score the test but also to score other booklets of PIRLS. Overall interrater reliability (across all booklets) was high (99% agreement) with PIRLS interrater reliability procedures being followed (Mullis et al., 2007). Children were awarded a score for each part of the test and a mean score for both parts (narrative max = 17, expository max = 16). Cronbach's alpha for this booklet was .72 for the total test.

Decoding Skills

Decoding skills were measured using the Three Minutes Test (Krom, Jongen, Verhelst, Kamphuis, & Kleintjes, 2010). This standardized test consists of three cards with lists of words differing in orthographic complexity. The first card has 150 high-frequency monosyllabic words of the type consonant-vowel-consonant (e.g., *cat*); the second card has 150 high-frequency monosyllabic words with consonant clusters at the beginning or end of the word (e.g., *spring*); and the third card has 120 less frequent polysyllabic words of increasing difficulty. For all three cards, the child must read as many words as possible in one minute. The child's score on each card is a combined speed/accuracy score and consists of the number of words read correctly in one minute. We transformed the scores on each of the three cards into z-scores and then calculated a mean z-score over the three cards. These mean z-scores were linearly transformed to the scale of Card 3 such that the mean and standard deviation at each measurement time was comparable to the mean and standard deviation of Card 3

for the complete sample. The reliability of the sum score of three cards was: Cronbach's alpha Grade 3 = .95, Grade 4 = .94 (Krom et al., 2010).

Vocabulary

Vocabulary was measured using a standardized test developed by Verhoeven and Vermeer (1999). In Grade 3, we used the third-grade test as the pretest and posttest, and in Grade 4, we used the fourth-grade test. Both tests consisted of 30 multiple-choice items. For each item, a sentence was presented in which one word was underlined. Students had to identify the meaning of this word from four alternatives. Cronbach's alpha was .85 for the third-grade test and .87 for the fourth-grade test. The scoring of these tests was based on Item Response Theory (IRT) to achieve comparability across grades (Grade 3 scale minimum = 9, maximum = 129; Grade 4: 37–159).

Nonverbal Intelligence

Nonverbal intelligence was measured using the Raven's Standard Progressive Matrices (Raven, Court, & Raven, 1977), a test comprised of five subsets. Subsets A to D (48 items) were administered as a measure of nonverbal intelligence and general reasoning skills. The test consisted of items presenting a matrix of geometric figures in which one item was missing. The correct missing entry must be selected from a set of answer choices. A reliability of .83 (Cronbach's alpha) is reported in the manual.

Procedure

Test Administration

All tests were administered by trained student assistants (third- and fourth-year students of psychology or special education), with the exception of one part of the PIRLS test, which was administered by the teachers.

Tests were administered according to a fixed schedule on two separate mornings, one week apart. On the first morning, the PIRLS (part 1, narrative text) was administered to the whole class and the Three Minutes Test was administered individually to each child. One week later, the questionnaire on reading strategies, the Reading Comprehension Test, and the vocabulary test were administered. During the week between these two sessions, the teachers administered the second part of the PIRLS test (part 2, expository text), for which they only had to repeat a short instruction. In January of Grade 4, the Raven's test was administered. Children who had moved to another school did not participate in the Raven's test and therefore have no scores recorded.

Statistical Analyses

A multilevel analysis was used to estimate the effects of the intervention. The data were hierarchically ordered for students within classrooms/teachers within schools, thus resulting in dependency in the data that can be dealt with by multilevel analysis (Snijders & Bosker, 2012). Furthermore, the multilevel model addresses variables at different levels. For instance, educational level of parents is both a student characteristic

and a classroom composition characteristic. Moreover, there were repeated measures for each student, treated as nested within students. The analyses were performed with MLwiN 2.22 (Rasbash, Charlton, Browne, Healy, & Cameron, 2009). See Appendix C in the online supplemental material for the treatment of missing values.

Results

Table 3 provides an overview of the means and standard deviations of the three measurement waves for the experimental group and the control group. The number of students differs per test because not all students were present at the time of each testing. The table reveals that on the pretest the experimental group had lower mean scores for reading comprehension, knowledge of reading strategies, vocabulary, and decoding skills than the control group (see Appendix C in the online supplemental material for an analysis of the pretest differences).

Multilevel Models

Within our data set, four levels are distinguished: measurement, student, classroom, and school. A first analysis reveals that the school level should be excluded because, in most cases, there were only one or two classrooms involved per school. Thus between-schools

Table 3. Means and standard deviations of the control and experimental group for all tests at the three measurement waves.

	Control			Experimental		
	<i>N</i>	<i>Mean</i>	<i>SD</i>	<i>N</i>	<i>Mean</i>	<i>SD</i>
Knowledge of Reading Strategies (max = 50)						
Pre	543	31.55	(4.821)	734	30.40	(4.931)
Mid post	543	34.28	(4.273)	736	35.51	(5.134)
Post test	519	36.52	(4.151)	679	38.30	(4.242)
Reading Comprehension Test (max = 29)						
Pre	536	14.61	(5.130)	718	13.28	(4.859)
Mid post	536	17.69	(5.692)	728	16.30	(5.449)
Post	517	20.53	(5.376)	674	19.87	(5.465)
PIRLS: Mean score						
Pre	556	6.390	(2.833)	734	5.74	(2.657)
Mid post	535	8.144	(2.950)	749	7.34	(2.985)
Post test	520	9.864	(2.900)	698	9.55	(3.031)
PIRLS: Narrative text (max = 17)						
Pre	542	7.17	(3.229)	723	6.40	(3.014)
Mid post	523	8.76	(3.100)	737	8.11	(3.232)
Post test	517	10.73	(3.107)	687	10.48	(3.239)
PIRLS: Expository text (max = 16)						
Pre	528	5.62	(2.933)	691	5.03	(2.804)
Mid post	514	7.60	(3.392)	714	6.56	(3.351)
Post test	507	8.96	(3.359)	676	8.59	(3.447)
Vocabulary (IRT-scaled score)						
Pre	526	81.27	(10.214)	724	79.25	(10.230)
Mid post	536	89.99	(11.411)	717	87.29	(12.546)
Post test	516	97.04	(11.793)	675	93.61	(11.039)
DMT Mean score						
Pre	562	49.62	(19.286)	726	47.33	(18.631)
Mid post	562	64.12	(18.272)	750	62.29	(18.899)
Post test	527	74.83	(17.801)	693	74.03	(18.052)
Raven (set A to D, max. = 48)	506	36.78	(5.626)	688	36.36	(5.830)

variance could hardly be distinguished from between-classrooms variance. By leaving out the school level, the variance between classrooms includes the school variance as well. As a consequence, the clustering effect was not fully accounted for in the estimates of the standard errors of the intervention effects, possibly leading to a (slight) underestimation of these standard errors and thus an overestimation of the *z*-statistics. As most schools had only one or two classes for each grade level, overestimation of the test statistics and disaggregation bias can be expected to be negligible. The clustering effect was taken into account by having classrooms as the highest level in the multilevel analyses. The difference between the intervention and the control condition was in the analyses as a variable at the classroom level. Therefore, the levels in the analyses did not completely match with the level at which randomization took place.

The three repeated measurements are represented in the model by two dummy variables to examine the intervention effect after one year and after two years, representing the differences with respect to the pretest. Because two dummy variables fully exhaust the differences between three measurements, there is no variance at the lowest level. In the end, the models reflected the variance in learning gains at the level of the students and at the level of the classrooms (including schools). The intervention effects were represented in the models by the coefficients for product variables between condition (intervention versus control) and each of the two dummy variables for measurement wave. A positive intervention effect implies that, on average, the learning gain on the first or second posttest is greater in classrooms from experimental schools than in classrooms from control schools when controlling for background variables.

Separate analyses were conducted for each dependent variable, that is, knowledge of strategies and reading comprehension. For reading comprehension, we used the scores on the Reading Comprehension Test as well as the scores on the PIRLS test. For each dependent variable, the results of the four models are presented. Model 1 shows the mean learning gain and the variances in the learning gain at both the student and classroom levels. In Model 2, the intervention effects were tested while controlling for the background variables—gender, age, SES, ethnicity/HL—at the student level. For the three categories of SES, we used two dummy variables, namely, SES low and SES lower middle, to represent low versus not disadvantaged and lower middle versus not disadvantaged. Ethnicity and HL (Ethn + HL, see [Table 1](#)) were represented by two dummy variables—non-Dutch and mixed. As a reference category, we used Dutch, meaning Dutch ethnicity of the parents, and Dutch HL. Age of the students was determined at the time of the pretest, that is, at the beginning of third grade (September 2009) and was expressed in years and centered by subtracting the sample mean. In Model 3, we added variables at the classroom level, i.e., the background variables of students aggregated by classrooms. The classroom means of the dummy variables are equal to proportions of the particular category in the classroom. With this model, we examined whether relations of the outcomes with background variables differ between individual and classroom levels. Model 4 is considered our final model and answers the question of whether the intervention had a positive effect on the criterion variables. Model 4 is more parsimonious than Model 3 in that all nonsignificant effects of background variables were left out. This model was also used to calculate standardized effect sizes for the intervention effect on the outcome variables.

To answer the question of differential effects, we added interaction effects to the model. In the next section, we first present the results of Models 1 to 4 for each outcome variable, after which the interaction effects are described.

Effects on Knowledge of Reading Strategies

As reading strategies were explicitly taught in the intervention group, this variable was closely related to the intervention. The results of the multilevel analyses are presented in Table 4. To explain the model parameters and the tables, the models are more extensively discussed for this first outcome variable.

Model 1 shows the learning gains for the entire sample. The intercept represents the mean score, 30.792, on the reading strategies questionnaire for the whole sample on the pretest. The coefficient of 4.046 for change by the end of Grade 3 indicates a mean learning gain of 4.046 points compared with the pretest. At the end of Grade 4, the mean learning gain was 6.546 compared with the pretest. The additional mean learning gain in Grade 4 was, accordingly, $6.546 - 4.046 = 2.5$. The numbers in brackets are the standard errors. Statistically

Table 4. Multilevel modeling results for knowledge of reading strategies.

	Model 1		Model 2		Model 3		Model 4	
	Coeff.	(SE)	Coeff.	(SE)	Coeff.	(SE)	Coeff.	(SE)
<i>Regression coefficients</i>								
Intercept (pretest)	30.792	(0.308)	31.243	(0.441)	31.358	(0.822)	30.968	(0.422)
Change by end Grade 3	4.046	(0.293)	2.676	(0.377)	2.672	(0.377)	2.673	(0.377)
Change by end Grade 4	6.546	(0.342)	4.906	(0.434)	4.901	(0.433)	4.907	(0.433)
Pretest difference			−1.111	(0.560)	−1.083	(0.554)	−1.096	(0.552)
SRI × Change end Grade 3			2.471	(0.505)	2.477	(0.505)	2.476	(0.505)
SRI × Change end Grade 4			2.946	(0.582)	2.942	(0.581)	2.951	(0.581)
Girls			1.461	(0.173)	1.470	(0.174)	1.494	(0.173)
Age (in years)			−0.849	(0.184)	−0.873	(0.186)	−0.875	(0.184)
Student SES: Lower middle			−1.034	(0.329)	−0.888	(0.334)	−0.914	(0.334)
Student SES: Low			−0.985	(0.262)	−0.883	(0.269)	−0.973	(0.265)
Ethn+HL: Mixed			−0.294	(0.236)	−0.196	(0.251)		
Ethn+HL: Non-Dutch			−0.706	(0.303)	−0.545	(0.320)		
Prop. girls in class					0.801	(2.100)		
Mean age of class					1.194	(1.343)		
Class SES: Lower middle					−4.758	(2.033)	−4.392	(1.937)
Class SES: Low					−0.361	(2.235)	−1.610	(0.895)
Class Ethn + HL: Mixed					0.174	(1.175)		
Class Ethn + HL: Non-Dutch					−1.459	(2.191)		
<i>Variances</i>								
<i>Classroom level</i>								
Intercept	4.909	(1.055)	3.856	(0.858)	3.615	(0.818)	3.625	(0.822)
Change by end Grade 3	4.127	(0.952)	2.700	(0.697)	2.693	(0.698)	2.691	(0.698)
Change by end Grade 4	5.850	(1.296)	3.813	(0.925)	3.803	(0.926)	3.803	(0.926)
<i>Student level</i>								
Intercept	19.532	(0.790)	18.356	(0.748)	18.359	(0.749)	18.376	(0.749)
Change by end Grade 3	22.855	(0.954)	22.839	(0.954)	22.839	(0.954)	22.835	(0.954)
Change by end Grade 4	24.207	(1.059)	24.080	(1.051)	24.068	(1.050)	24.091	(1.051)
−2 loglikelihood:	21,364.029		21,003.327		20,994.976		20,999.757	
N classrooms	62		62		62		62	
N students	1,448		1,423		1,423		1,423	
N scores	3,753		3,722		3,722		3,722	

Note. Estimates of covariances omitted. Statistically significant results ($p < .05$) are in boldface.

significant effects ($p < .05$) are in bold. There was a statistically significant average learning gain both at the end of Grade 3 and at the end of Grade 4 compared with the pretest scores at the beginning of Grade 3.

Each model has variance components at the student and the classroom levels. The variance of the student-level intercept represents individual differences on pretest scores of students within classrooms. The variances of the dummy variables for change by the end of Grade 3 and by the end of Grade 4 represent individual differences in learning gains of students during Grade 3, respectively during the two-year period for Grades 3 and 4. Model 1 shows statistically significant differences between classrooms in achievement as well as in learning gain after one year, at the end of Grade 3, and after two years, at the end of Grade 4. Note that no distinctions have yet been made between experimental and control schools.

In addition to the variances, there were three covariances at each level, which are not presented in the tables. In general, there was a negative correlation between learning gain and score on the pretest, indicating the higher the score on the pretest, the smaller the learning gain. The learning gain at the end of Grade 3 correlated positively with the learning gain at the end of Grade 4. At the group level, for most of the outcome variables, there were no statistically significant correlations, other than that for the reading strategies questionnaire. A comparison of the intercept variances on the pretest at the student and group levels revealed that 20% of the variance was related to differences between classrooms (and schools). In other words, the intra-class correlation was .20.

In Model 2, we tested the effects of the intervention while controlling for background variables. At the end of third grade, there was a statistically significant difference in mean learning gain between the experimental and the control group in favor of the experimental group (SRI End Grade 3: $b = 2.471$, $z = 4.893$, $p < .001$). At the end of Grade 4, this intervention effect had further increased. Accordingly, after two years of intervention, the mean difference in learning gain on the reading strategies questionnaire between the experimental group and the control group was 2.946 in favor of the experimental group. On the pretest, the experimental group lagged behind the control group (labeled in the tables as “Pretest difference,” $b = -1.111$, $z = 1.983$, $p = .047$), but they caught up during the intervention period. With respect to the background variables, the model showed statistically significant differences on the pretest for gender (girls obtained higher scores than boys), age (younger students obtained higher scores than older students), SES (students from families with higher educational level performed better) and ethnicity/HL (students with a non-Dutch background had, on average, poorer results than the other two categories). By adding the background variables to the model, the intercept variance at the classroom level decreased significantly, but at the individual level, the variance decreased only minimally. In Model 2, the classroom-level variance of the learning gain was reduced by 35% both after one and after two years of intervention. In Model 3, we added the background variables at the classroom level to examine whether classroom composition characteristics such as gender, age, SES, and ethnicity/HL had additional effects on the test scores at baseline. In other words, we tested whether background variables had the same effect at the classroom level as they did at the individual level. This turned out to be the case for gender, age, and ethnicity/HL, as shown by nonsignificant effects for these aggregated variables. For classroom SES, however, there appeared to be an amplified negative effect for the proportion of students in the lower middle category. There was, however, no such effect for the proportion of students with the lowest parental educational level. By adding the effects of classroom composition variables to

the model, there was no significant decrease in the deviance ($-2 \times \text{loglikelihood}$) of the model compared with Model 2, $\chi^2(6) = 8.351$, $p = .214$. Furthermore, the estimation of the intervention effect was not importantly affected by the presence or absence of classroom composition variables.

Finally, Model 4 in Table 4 is a more parsimonious version of Model 3, as all nonsignificant effects of background variables are excluded. The fit of the model was not significantly affected $\chi^2(6) = 4.781$, $p = .572$ (comparison of the deviances of Model 4 and Model 3). Model 4, as the final model, is the basis on which we can answer our main question. This model is also used to calculate the effect sizes. Based on Model 4, we conclude that the intervention did indeed lead to increased knowledge of reading strategies. The effect proved to be stable over Models 2, 3, and 4.

Standardized effect sizes were calculated by dividing the mean difference in learning gain at the end of Grade 3 and at the end of Grade 4 by the standard deviation derived from the intercept variance at the student level from Model 1. The effect sizes for knowledge of reading strategies were moderate (end of Grade 3: $d = 0.56$; end of Grade 4: $d = 0.67$).

Effects on Reading Comprehension

For reading comprehension, we examined the results of the Reading Comprehension Test of Aarnoutse and Schellings (Table 5) and of the PIRLS test (Table 6).

Reading Comprehension Test

Model 1 shows that the learning gains on the Reading Comprehension Test at the end of Grade 3 and at the end of Grade 4 compared with the pretest are statistically significant. At the end of Grade 3, the mean test score was raised by 2.955, and at the end of Grade 4, the mean test score increased by 6.107. Model 1 further shows statistically significant differences in test scores between classrooms as well as in learning gains between classrooms over time. Much larger, however, were the individual differences between students within classrooms both with respect to pretest achievement and learning gains. Based on the variances of the intercept, the intra-class correlation on the pretest was .19, which means that 19% of the total variance in test scores was related to differences between classrooms (or schools).

In Model 2, the intervention effects on the Reading Comprehension Test were examined, while controlling for the background variables. At the end of Grade 3, there was no statistically significant difference in learning gain between the two conditions ($b = 0.210$, $z = 0.78$, $p = .437$), but after two years, at the end of Grade 4, students in the experimental condition showed a statistically significant higher learning gain ($b = 0.914$, $z = 2.59$, $p = .010$). After two years, the mean difference in learning was 0.914 in favor of the intervention group. On the pretest, the children in the intervention groups had a statistically significant lower mean score than the children in the control groups ($b = -1.011$). In other words, the intervention group scored approximately one point lower than the control group on the pretest, but after two years of intervention, the intervention group scored, on average, almost one point higher than the control group.

With respect to the background variables, the same pattern was found as that for the Knowledge of Reading Strategies questionnaire. The model showed statistically significant differences on the pretest for gender (girls obtained, on average, higher scores than boys), age (younger students obtained higher scores than older students), SES (students with higher

Table 5. Multilevel modeling results for the Reading Comprehension Test.

	Model 1		Model 2		Model 3		Model 4	
	Coeff.	(SE)	Coeff.	(SE)	Coeff.	(SE)	Coeff.	(SE)
<i>Regression coefficients</i>								
Intercept (pretest)	13.704	(0.302)	15.281	(0.372)	15.120	(0.765)	14.852	(0.369)
Change by end Grade 3	2.955	(0.134)	2.849	(0.204)	2.867	(0.204)	2.863	(0.204)
Change by end Grade 4	6.107	(0.181)	5.607	(0.264)	5.645	(0.262)	5.645	(0.263)
Pretest difference			−1.011	(0.437)	−0.716	(0.410)	−0.803	(0.408)
SRI × Change end Grade 3			0.210	(0.270)	0.193	(0.271)	0.195	(0.270)
SRI × Change end Grade 4			0.914	(0.353)	0.881	(0.350)	0.885	(0.351)
Girls			0.577	(0.222)	0.612	(0.223)	0.595	(0.222)
Age (in years)			−1.711	(0.235)	−1.754	(0.238)	−1.710	(0.234)
Student SES: Lower middle			−1.386	(0.413)	−1.115	(0.422)	−1.114	(0.422)
Student SES: Low			−1.415	(0.332)	−1.111	(0.343)	−1.107	(0.342)
Ethn + HL: Mixed			−1.376	(0.298)	−1.050	(0.322)	−0.997	(0.310)
Ethn + HL: Non-Dutch			−1.985	(0.378)	−1.479	(0.405)	−1.528	(0.397)
Prop. girls in class					−0.566	(2.101)		
Mean age of class					1.094	(1.351)		
Class SES: Lower middle					−5.124	(2.048)	−4.983	(1.967)
Class SES: Low					−2.469	(2.194)	−3.925	(0.995)
Class Ethn + HL: Mixed					0.617	(1.180)		
Class Ethn + HL: Non-Dutch					−2.691	(2.150)		
<i>Variances</i>								
<i>Classroom level</i>								
Intercept	4.652	(1.015)	1.932	(0.521)	1.446	(0.433)	1.477	(0.439)
Change by end Grade 3	0.324	(0.199)	0.316	(0.199)	0.317	(0.199)	0.315	(0.199)
Change by end Grade 4	0.946	(0.362)	0.813	(0.337)	0.789	(0.333)	0.794	(0.333)
<i>Student level</i>								
Intercept	20.368	(0.824)	19.629	(0.800)	19.626	(0.800)	19.627	(0.800)
Change by end Grade 3	15.203	(0.642)	15.222	(0.645)	15.218	(0.645)	15.219	(0.645)
Change by end Grade 4	19.171	(0.852)	19.115	(0.850)	19.124	(0.851)	19.114	(0.850)
−2 loglikelihood:	20,890.793		20,565.843		20,542.421		20,546.353	
N classrooms	62		62		62		62	
N students	1,443		1,419		1,419		1,419	
N scores	3,708		3,678		3,678		3,678	

Note. Estimates of covariances omitted. Statistically significant results ($p < .05$) are in boldface.

educated parents performed better) and ethnicity/HL background (students with non-Dutch background had, on average, poorer results than the other two categories). By adding the background variables to the model, the intercept variance at the group level strongly decreased, while it decreased only minimally at the individual level. In Model 2, the variance in the learning gain at the classroom level was reduced by 14.1% after two years of intervention.

In Model 3, we added the background variables at the classroom level. Again, only the classroom proportion of students from the lower middle category showed a statistically significant and negative effect. The classroom composition variables as a whole showed a statistically significant contribution to the model compared with Model 2, $\chi^2(6) = 23.42$, $p < .001$. However, the estimation of the intervention effect was not importantly affected by the classroom composition variables, and the conclusions remained the same as those for Model 2.

Finally, Model 4 shows that leaving out nonsignificant effects did not change the fit of the model, $\chi^2(4) = 3.93$, $p = .416$ (comparison of deviances for Models 3 and 4). Model 4 leads us to the conclusion that after two years, the intervention had a significant positive impact on reading comprehension results as measured by the Reading Comprehension Test. After

Table 6. Multilevel modeling results for the PIRLS test (mean score).

	Model 1		Model 2		Model 3		Model 4	
	Coeff.	(SE)	Coeff.	(SE)	Coeff.	(SE)	Coeff.	(SE)
<i>Regression coefficients</i>								
Intercept (pretest)	5.899	(0.187)	6.794	(0.219)	6.785	(0.387)	6.473	(0.199)
Change by end Grade 3	1.694	(0.082)	1.737	(0.126)	1.739	(0.125)	1.739	(0.126)
Change by end Grade 4	3.591	(0.105)	3.360	(0.153)	3.371	(0.153)	3.374	(0.154)
Pretest difference			−0.486	(0.264)	−0.283	(0.213)	−0.317	(0.221)
SRI × Change end Grade 3			−0.067	(0.167)	−0.067	(0.166)	−0.068	(0.166)
SRI × Change end Grade 4			0.428	(0.205)	0.428	(0.205)	0.430	(0.206)
Girls			0.280	(0.119)	0.296	(0.120)	0.299	(0.119)
Age (in years)			− 0.911	(0.127)	− 0.931	(0.128)	− 0.921	(0.126)
Student SES: Lower middle			− 0.482	(0.223)	−0.313	(0.227)	−0.308	(0.227)
Student SES: Low			− 0.744	(0.179)	− 0.519	(0.184)	− 0.510	(0.184)
Ethn+HL: Mixed			− 0.867	(0.162)	− 0.623	(0.173)	− 0.599	(0.165)
Ethn+HL: Non-Dutch			− 1.216	(0.204)	− 0.841	(0.217)	− 0.888	(0.212)
Prop. girls in class					0.916	(1.054)		
Mean age of class					0.253	(0.682)		
Class SES: Lower middle					− 3.216	(1.033)	− 3.166	(1.007)
Class SES: Low					−1.647	(1.107)	− 3.036	(0.513)
Class Ethn+HL: Mixed					0.272	(0.593)		
Class Ethn+HL: Non-Dutch					− 2.145	(1.084)		
<i>Variances</i>								
<i>Classroom level</i>								
Intercept	1.886	(0.389)	0.791	(0.190)	0.387	(0.117)	0.453	(0.129)
Change by end Grade 3	0.199	(0.075)	0.202	(0.075)	0.199	(0.075)	0.199	(0.075)
Change by end Grade 4	0.392	(0.121)	0.350	(0.113)	0.352	(0.114)	0.358	(0.115)
<i>Student level</i>								
Intercept	5.784	(0.231)	5.542	(0.223)	5.524	(0.222)	5.523	(0.222)
Change by end Grade 3	4.201	(0.175)	4.199	(0.176)	4.198	(0.176)	4.198	(0.176)
Change by end Grade 4	5.049	(0.221)	5.073	(0.222)	5.072	(0.222)	5.071	(0.222)
−2 loglikelihood:	16,569.301		16,262.391		16,217.265		16,223.123	
N classrooms	62		62		62		62	
N students	1,461		1,433		1,433		1,433	
N scores	3,789		3,753		3,753		3,753	

Note. Estimates of covariances omitted. Statistically significant results ($p < .05$) are in boldface.

one year of intervention, however, this was not yet the case. The standardized effect sizes were 0.04 for the change in Grade 3 and 0.20 at the end of Grade 4.

PIRLS Test

In Table 6, Model 1 reveals the same results for the PIRLS test as Model 1 did for the reading strategies questionnaire and the Reading Comprehension Test, that is, there was a significant mean learning gain over time and a statistically significant variance in learning gain at the classroom level. The intra-class correlation for the pretest was .25.

In Model 2, the model with the intervention effects controlled for background variables, there was no statistically significant intervention effect at the end of Grade 3 ($b = -0.067$, $z = -0.40$, $p = .688$), but at the end of Grade 4, a statistically significant and positive intervention effect showed up ($b = 0.428$, $z = 2.088$, $p = .037$). The students in the intervention schools, on average, caught up with students of the control schools after two years of SRI intervention. With respect to the background variables, the same pattern as for the previous criterion variables was found. The model showed statistically significant differences on the pretest for gender (girls obtained, on average, higher scores than boys), age (younger

students obtained higher scores than older students), SES (students with better educated parents performed better) and ethnicity/HL (Dutch students performed, on average, better than students with non-Dutch background). By adding the background variables to the model, the intercept variance strongly decreased, but only at the classroom level. In Model 2, the classroom-level variance in learning gains decreased by 11% after two years of intervention.

When the background variables at the group level were added in Model 3, gender, age, and ethnicity/HL factors had no additional effect. However, for the students with a lower middle or a low educational background, there was an amplified negative effect of classroom composition. By adding the effects of group composition variables to the model, there was a statistically significant decrease of the deviance in comparison with Model 2, $\chi^2(6) = 45.13$, $p < .001$. The estimation of the intervention effect, however, was, again, not affected.

Finally, Model 4 indicates that after omitting the nonsignificant effects of background variables, the fit of the model did not change in comparison with Model 3, $\chi^2(4) = 5.858$, $p = .21$, and the estimates of the intervention effect changed only marginally. Accordingly, with respect to the mean score on the PIRLS test, positive effects of the intervention were found after a two-year intervention period ($b = 0.43$, $z = 2.09$, $p < .05$, $d = 0.18$).

In additional analyses the two subtests of the PIRLS, narrative and expository, were separately analyzed. The results of these analyses can be found in Appendix C of the online supplemental material.

Differential Effects of the Intervention

To answer our second research question, that is, are there differential effects of the intervention, we examined whether the intervention effects differed for students with different values on the background variables (SES, ethnicity/HL, age) and whether the intervention effects were affected by some covariates that were not yet in the models, such as nonverbal intelligence, vocabulary knowledge and decoding skills. SES and ethnicity/HL were represented in the models by two dummy variables each. To test the interactions of these two background variables with the intervention effects, we added 20 product variables to Model 3 from Tables 4 to 6.

Because SES and ethnic/HL background were highly correlated, we aggregated these variables into one variable by means of principal components analysis for nominal and ordinal data (Gifi, 1990). The analysis was performed on the original categories of SES, ethnicity, and home language, which resulted in a principal component with a Cronbach's alpha of .805. This principal component was added to Model 3 instead of the four dummy variables. Thus, in the model with the interactions at the student level, five product variables were now needed rather than 20. No statistically significant interactions of the background variables with the intervention effect were found, nor were any significant interactions found when the dummy variables were replaced by the principal component (not shown in the tables). The conclusion is that no evidence of differential intervention effects was found with respect to SES or ethnicity and home language.

We then examined whether the intervention effects were moderated by the age of the children. This was tested by adding product variables with age to Model 3 for all criterion variables. We also replaced the dummy variables for SES and ethnicity/home language by the principal component in these models, and again, no statistically significant interaction with

the intervention effect was found. Finally, we added three covariates to Model 4: Raven's nonverbal intelligence, the vocabulary test, and the decoding test. For the Raven's test, one measurement was available, whereas there were time-varying covariates measured at each of the three occasions for vocabulary and decoding. It is further important to note that, with respect to the Raven, due to numerous missing values, the analyses were performed on the data of a smaller sample.

We first tested mean differences on the Raven between the two experimental conditions, and no significant differences were found ($M_{\text{contr}} = 36.78$, $SD = 5.63$; $M_{\text{exp}} = 36.36$, $SD = 5.97$; $t(1191) = 1.121$, $p = .23$). The Raven correlated positively with all variables in the same way for both conditions. For the reading comprehension tests, it was found that the higher the score on the Raven, the larger the learning gain. No relation was found, however, with learning gain on the reading strategies questionnaire. No significant interaction between the Raven and the intervention effect was found for any of the criterion variables. Thus, there is no evidence of differential effects of the intervention for children with higher or lower intelligence. The intervention effect at the end of Grade 4 on the PIRLS test was no longer statistically significant after controlling for the Raven ($z = 1.89$, $p = .059$).

Vocabulary, as a time-varying covariate, was significantly and positively related with all criterion variables. These relations were the same for the experimental and the control group. Inconsistent relations, however, with learning gain were found. For the reading comprehension tests no relations with learning gain were found but negative relations appeared for the questionnaire on reading strategies. Controlling for vocabulary did not lead to different conclusions regarding the effects of the intervention; the estimated effect size of the intervention, though, increased for all reading comprehension tests. There were no statistically significant interactions between the intervention effect and the level of vocabulary.

Decoding skills, also a time-varying covariate, had a statistically significant positive relationship with all criterion variables for both conditions. But learning gain was not related to the level of decoding skills. As with the Raven, the intervention effect on the PIRLS test at the end of Grade 4 was no longer significant ($z = 1.714$, $p = .086$). More important, there were no significant interactions between decoding skills and the intervention effect. Thus, again, there was no evidence of differential intervention effects.

Discussion

The effects of an approach for sustained strategic reading comprehension instruction in third and fourth grades on knowledge of strategies and reading comprehension were examined in intermediate elementary classrooms. The role of student characteristics was taken into account. Several conclusions can be drawn from the results of the study. We found a clear positive effect on student strategy knowledge. At the end of Grade 3, after one year of intervention, and at the end of Grade 4, after two years of intervention, the students in the intervention group exhibited greater growth with respect to knowledge of reading strategies than the control group. Gaining an understanding of reading strategies was a core element of the approach. Thus, it is concluded that the intervention was effective in enhancing children's declarative knowledge of these strategies. It must be kept in mind that the teachers of the control group also used programs aimed at the learning of reading strategies. Despite this, at the end of Grade 3, after one year of intervention, and at the end of Grade 4, after two years of intervention, the students in the intervention group exhibited greater growth

with respect to knowledge of reading strategies than the control group. The effect size was substantial; compared with an average student of the control group, the average student in the experimental condition improved by 21 percentile points already at the end of grade 3 and by 25 percentile points at the end of grade 4.

With respect to reading comprehension, at the end of Grade 4 significant positive effects were evidenced on the two general reading comprehension tests. The effect sizes were small, but consistent with other studies (e.g., Souvignier & Mokhlesgerami, 2006; Spörer et al., 2009). The improvement index in reading comprehension was about 8 percentile points at the end of Grade 4, while hardly any improvement was evidenced at the end of Grade 3. The results for the PIRLS test showed a stronger intervention effect for the narrative text than for the expository text (see Appendix D in the online supplemental material).

Our findings suggest that Grade 3 children should first enhance their declarative knowledge of reading strategies. They must learn what reading strategies are, and through the modeling behavior of the teacher, they then should learn how the reading strategies can be used and verbalized. Second, the children must learn to apply this knowledge when reading texts; the more the child uses the strategies, the more internalized the strategies become as a transfer from declarative to procedural knowledge occurs over time. These findings corroborate earlier studies of Davis (2010), Rosenshine and Meister (1994), and Souvignier and Mokhlesgerami (2006) and fit the models of self-regulated learning, which stress that the modeling of strategies by the teacher lead to the internalization of these strategies by students who then apply the strategies when reading text independently. So, the modeling of strategies in the context of reading aloud and the practice of students to apply these strategies during independent reading may be the effective key components in the SRI program. It is probable that the focus on teacher modeling and on the verbalization of strategy use provided students the opportunity to build declarative knowledge regarding the reading strategies in Grade 3 and that the continued modeling by the teachers and the students in Grade 4 helped the students to internalize the procedural use of the strategies when independently reading text.

With respect to our second research question, we concluded that no differential effects of the intervention were found. This means that the effects found with respect to knowledge of reading strategies and reading comprehension were general effects that apply to the intervention group in general, irrespective of the children's background characteristics, decoding skills, or vocabulary knowledge. Accordingly, we concluded that the SRI program is a robust program that enhances children's reading comprehension.

From our observations of the read-aloud and independent reading lessons, we determined that most teachers conducted the lessons as prescribed. Thus, the short training with video clips was deemed effective. The scripted instructions of the modeling of strategies benefited the teachers in their implementation of the lessons and in their understanding of the meaning and intentions of the program. In the conversations we had with the teachers, they mentioned that they had become more proficient and confident in modeling the strategies and that they applied this new ability to other domains and subjects. For further implementation of the SRI program, the scripted instruction may be of help when teachers start to implement this kind of instruction during reading aloud. The experience with the scripts may help the teachers to construct modeling prompts themselves when choosing their own books.

The present study has some limitations. The study is a first step in identifying the benefits of a sustained reading program in the natural classroom setting. We focused on the effects of the program on children's knowledge of strategies and reading comprehension. The SRI program involved nine strategies. In the present study, we were not able to specify whether a particular strategy was effective. Future research should include more refined measures of self-regulation, meta-cognitive knowledge and skills, and measures of motivation to gain a better understanding of the influence of the strategies on reading comprehension instruction.

Another issue in this study is that we conducted only two classroom observations, one per grade in addition to the training sessions. During the observations we found that the lessons were conducted as intended. As the role of the teacher in the SRI program is crucial, a much more in-depth knowledge of teacher behavior is needed to better understand the degree to which modeling enhances reading comprehension and the way it helps children in applying these strategies in narrative and expository text. More thorough fidelity data could then be included in the statistical analyses. In such a study, the focus on differentiation in modeling and scaffolding of strategies for students with varying reading abilities should be considered to develop an approach that will best meet the individual needs of the children. In this approach a stronger differentiation in the instruction of strategies in narrative and expository text may help to gain insight into the effects of strategy instruction on these different types of texts.

To conclude, the present study underlines the importance of long-term intervention studies in natural classrooms. It is because we had the possibility to conduct a two-year study that we found effects on reading comprehension. The SRI approach does enhance the knowledge of strategies and reading comprehension among children enrolled in the third and fourth grades. Such an approach may bridge the step from declarative knowledge to procedural knowledge, which is needed to apply strategies effectively when reading text. For teachers, it is important that they know that the modeling of strategies should be embedded in a *sustained* approach of reading comprehension instruction, as they guide children to become independent readers.

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Appendix: Examples of Children's Books Used in Intervention

- Menten, T. (2008). *Juffrouw Pots*. Houten, the Netherlands: Van Goor.
- Oldenhave, M. (2010). *Klem!* Amsterdam, the Netherlands: Ploegsma.
- Rood, L. (2010). *Overleven op drakeiland*. Amsterdam, the Netherlands: Leopold.